

1 Prime Factorization

Prime factorization is the process of expressing a number as the product of its prime factors. Prime numbers are numbers greater than 1 that have no divisors other than 1 and themselves.

Key Concepts

- **Fundamental Theorem of Arithmetic:** Every integer greater than 1 is either a prime number or can be expressed as a product of prime numbers, uniquely (apart from the order of the factors). - **Finding Prime Factors:** Repeatedly divide the number by the smallest prime until the quotient is 1.

Example Explanation

Consider the number 60. To find its prime factorization:

- Divide by 2: $60 \div 2 = 30$.
- Divide by 2 again: $30 \div 2 = 15$.
- Divide by 3: $15 \div 3 = 5$.
- 5 is prime, so the process stops.

$$60 = 2^2 \cdot 3 \cdot 5$$

Practice Problems

1. Find the prime factorization of 84.
2. Express 210 as a product of its prime factors.
3. Determine the prime factors of 96.
4. Write the prime factorization of 450.
5. What is the largest prime factor of 1001?

Solutions

1. $84 = 2^2 \cdot 3 \cdot 7$.
2. $210 = 2 \cdot 3 \cdot 5 \cdot 7$.
3. $96 = 2^5 \cdot 3$.
4. $450 = 2 \cdot 3^2 \cdot 5^2$.
5. $1001 = 7 \cdot 11 \cdot 13$, so the largest prime factor is 13.

2 Divisibility Rules

Divisibility rules are shortcuts to determine whether a number is divisible by another without performing long division.

Key Concepts

- **Divisibility by 2:** A number is divisible by 2 if its last digit is even.
- **Divisibility by 3:** A number is divisible by 3 if the sum of its digits is divisible by 3.
- **Divisibility by 5:** A number is divisible by 5 if its last digit is 0 or 5.
- **Divisibility by 6:** A number is divisible by 6 if it is divisible by both 2 and 3.
- **Divisibility by 9:** A number is divisible by 9 if the sum of its digits is divisible by 9.

Example Explanation

Consider the number 234. Check divisibility:

- Divisible by 2: The last digit is 4, which is even.
- Divisible by 3: The sum of the digits is $2 + 3 + 4 = 9$, which is divisible by 3.
- Divisible by 5: The last digit is not 0 or 5, so 234 is not divisible by 5.

234 is divisible by 2 and 3 but not 5.

Practice Problems

1. Is 126 divisible by 9?
2. Check if 3450 is divisible by 6.
3. Determine whether 721 is divisible by 3.
4. Is 128 divisible by 4?
5. Verify if 10010 is divisible by 5.

Solutions

1. Sum of digits of 126 is $1 + 2 + 6 = 9$, which is divisible by 9. So, 126 is divisible by 9.
2. 3450 is divisible by 2 (last digit is 0) and by 3 (sum of digits $3 + 4 + 5 + 0 = 12$ is divisible by 3). Hence, it is divisible by 6.
3. Sum of digits of 721 is $7 + 2 + 1 = 10$, which is not divisible by 3. So, 721 is not divisible by 3.
4. The last two digits of 128 are 28, which is divisible by 4. So, 128 is divisible

- by 4.
 5. The last digit of 10010 is 0, so it is divisible by 5.

3 LCM/GCD Properties

The Least Common Multiple (LCM) and Greatest Common Divisor (GCD) are fundamental concepts in number theory used to analyze relationships between integers.

Key Concepts

- **Definition of GCD:** The largest positive integer that divides two or more integers without leaving a remainder. - **Definition of LCM:** The smallest positive integer that is a multiple of two or more integers. - **Relationship between GCD and LCM:**

$$\gcd(a, b) \times \text{lcm}(a, b) = a \times b$$

- **Properties of gcd and lcm:** - $\gcd(a, b) = \gcd(b, a \bmod b)$ (Euclidean Algorithm) - $\text{lcm}(a, b) = \frac{|a \cdot b|}{\gcd(a, b)}$

Example Explanation

Consider $a = 12$ and $b = 15$:

- $\gcd(12, 15) = 3$ (largest divisor common to both).
- $\text{lcm}(12, 15) = \frac{12 \cdot 15}{3} = 60$.

Practice Problems

1. Find the gcd and lcm of 18 and 24.
2. Determine the lcm of 8, 12, and 16.
3. Compute the gcd of 81 and 27.
4. If $\gcd(a, b) = 4$ and $a \times b = 96$, find $\text{lcm}(a, b)$.
5. Use the Euclidean Algorithm to find $\gcd(56, 98)$.

Solutions

1. $\gcd(18, 24) = 6$, $\text{lcm}(18, 24) = 72$.
2. $\text{lcm}(8, 12, 16) = 48$.
3. $\gcd(81, 27) = 27$.
4. $\text{lcm}(a, b) = \frac{96}{4} = 24$.
5. Using the Euclidean Algorithm: $98 \bmod 56 = 42$, $56 \bmod 42 = 14$, $42 \bmod 14 = 0$, so $\gcd(56, 98) = 14$.

4 Modular Arithmetic

Modular arithmetic, also known as "clock arithmetic," is a system of arithmetic for integers where numbers wrap around upon reaching a certain value (the modulus).

Key Concepts

- **Modulus Operation:** $a \bmod n$ gives the remainder when a is divided by n .
- **Congruence:** $a \equiv b \pmod{n}$ means a and b leave the same remainder when divided by n .

Properties of Modular Arithmetic:

- $(a + b) \bmod n = [(a \bmod n) + (b \bmod n)] \bmod n$
- $(a - b) \bmod n = [(a \bmod n) - (b \bmod n)] \bmod n$
- $(a \cdot b) \bmod n = [(a \bmod n) \cdot (b \bmod n)] \bmod n$
- **Exponentiation:** $(a^k) \bmod n$ can be computed efficiently using modular exponentiation.

Example Explanation

Consider $a = 17$, $b = 23$, and $n = 5$:

- $17 \bmod 5 = 2$, $23 \bmod 5 = 3$.
- $(17 + 23) \bmod 5 = (2 + 3) \bmod 5 = 0$.
- $(17 \cdot 23) \bmod 5 = (2 \cdot 3) \bmod 5 = 6 \bmod 5 = 1$.

Practice Problems

1. Compute $29 \bmod 7$.
2. Solve $15x \equiv 5 \pmod{10}$ for x .
3. Simplify $(123 + 456) \bmod 11$.
4. Find $(7^4) \bmod 3$.
5. Determine if $68 \equiv 4 \pmod{16}$.

Solutions

1. $29 \bmod 7 = 1$.
2. $15x \equiv 5 \pmod{10}$ simplifies to $5x \equiv 5 \pmod{10}$, so $x \equiv 1 \pmod{2}$ (any odd number).
3. $(123 + 456) \bmod 11 = 579 \bmod 11 = 6$.
4. $(7^4) \bmod 3 = (49) \bmod 3 = 1$.
5. $68 \bmod 16 = 4$, so $68 \equiv 4 \pmod{16}$.